At an extremely small scale, space is a fluctuating swarm of quanta of gravity which act upon each other, and together act upon things, manifesting themselves in these interactions as spin networks, grains interrelated with each other (figure 6.5).

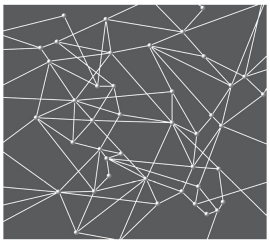


Figure 6.5 At a minute scale, space is not continuous: it is woven from interconnected finite elements.

Physical space is the fabric resulting from the ceaseless swarming of this web of relations. The lines themselves are nowhere; they are not in a place but rather create places through their interactions. Space is created by the interaction of individual quanta of gravity.

This is the first step towards understanding quantum gravity. The second concerns time. And, to time, the next chapter is devoted.

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* The result of the calculation is simple. I show it here so you can see how Dirac's spectra work. The possible values of the area *A* are given in the following formula, where *j* is a 'half-integer', that is to say, a number which is half of an integer, such as 0, ½, 1, $^3/_2$, 2, $^5/_2$, 3 …

$$A = 8\pi L_p^2 \sqrt{j(j+1)}$$

A is the area that a surface separating two grains of space can have. 8 is the number eight, nothing special about it. π is the Greek pi which we studied at school: the constant which gives the relation between the circumference and the diameter of any circle, and which appears everywhere in physics, I don't know why. *Lp* is the Planck length, the extremely small scale at which the phenomena of quantum gravity take place. L^2p is the square of *Lp*, which is the (extremely small) area of a tiny square with sides equal to the Planck length. Therefore $8\pi L^2p$ is simply a 'small' area: the area of a minuscule square with a side which is about a millionth of a billionth of a billionth of a billionth of a centimetre (10^{-66} cm²). The interesting aspect of the formula is the square root and what is within it. The key point is that *j* is a half-integer, that is to say, it may have only values which are multiples of ½. For each one of these, the root has a certain value, listed approximately in table 6.1.